Deep Learning

Sequence prediction architectures for financial streams

Project report

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Executive summary

Overview and objective

This research project investigates the predictive power of deep learning architectures in the context of financial market forecasting. Specifically, we developed an automated pipeline to extract the "Item 1A: Risk Factors" section from over 9,000 SEC 10-K filings (S&P 500 constituents) over a 20-year horizon (2005–2025). The objective was to determine if qualitative linguistic disclosures can predict a firm's binary stock performance over the subsequent 12-month fiscal period.

1.2 Methodological Approach

Thirteen distinct neural network architectures identify the optimal balance between sequential memory and feature extraction:

- **Recurrent Models**: Baseline, Stacked, Wider, and Bidirectional variants of both LSTM and GRU cells.
- **Attention Mechanisms:** Integration of custom Attention layers to weigh high-impact financial keywords within 3,000-token sequences.
- **Transfer Learning**: A hybrid FinBERT-Bidirectional GRU model leveraging domain-specific pre-trained embeddings.
- **Convolutional Models:** Single-layer, Deep Sequential, and Parallel Multi-Kernel CNN architectures designed for local and hierarchical n-gram pattern recognition.

Research significance

This project contributes to the evolving field of Financial Natural Language Processing (FinNLP) and engages with the ongoing academic debate surrounding the Efficient Market Hypothesis (EMH). By treating financial text not merely as documentation but as a complex sequence of informative signals, the study explores the capacity of advanced deep learning architectures to bridge the gap between unstructured narrative data and structured financial outcomes.

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Data acquisition and preprocessing methodology

This section delineates the technical pipeline developed to transform unstructured regulatory filings and historical market data into a machine-learning-ready dataset. The process encompasses automated web scraping, high-speed HTML parsing, and the synchronization of temporal financial metrics..

I. Automated data acquisition

To ensure a comprehensive representation of the U.S. equity market, the data acquisition strategy targeted the **S&P 500 index**.:

- **Universe definition**: The system dynamically retrieved the list of S&P 500 constituent tickers via Wikipedia to ensure the study reflected the most current market leaders.
- **SEC edgar integration**: Using the "sec_edgar_downloader" framework, the pipeline conducted a longitudinal search for **Form 10-K** filings.
- **Temporal boundaries**: The retrieval parameters were strictly defined to cover a twenty-year horizon, specifically fetching filings from **January 1, 2005, through December 31, 2025**.
- **Compliance and velocity**: To adhere to SEC "Fair Access" limits, the scraping engine implemented a 0.2-second delay between requests, successfully downloading over **9,165 primary documents**.

II. Textual extraction and noise reduction

The primary challenge in financial sequence prediction is the "signal-to-noise" ratio within SEC documents. Form 10-Ks are notoriously dense, often exceeding 100 pages of mixed formatting.

- **Targeted parsing**: The extraction engine utilized BeautifulSoup with the lxml parser to isolate **Item 1A: Risk Factors**.
- **Decomposition of non-textual elements**: To optimize computational efficiency and preserve sequence integrity, the algorithm programmatically "decomposed" (deleted) all HTML tables, images, and SVG graphics. This step ensures the model processes narrative sequences without being distracted by tabular financial data or visual artifacts.
- **Heuristic mapping**: Because SEC filings do not use standardized HTML IDs for sections, the pipeline employed **Regular Expressions (Regex)** to identify the start and end markers for Item 1A (e.g., searching for "Item 1A" while terminating at "Item 1B" or "Item 2").
- **Quality control (flagging)**: A validation layer was implemented to filter out unsuccessful extractions. This included identifying "Table of Contents" matches by enforcing a minimum character threshold (1,500 characters) for any extracted segment.

To facilitate supervised learning, each textual sequence was paired with a forward-looking financial performance metric. This ensures that the linguistic patterns found in the "Risk Factors" are directly mapped to tangible market outcomes.

Market data integration

- **Data retrieval**: Historical price data was fetched via the yahoo finance (yfinance) API for every ticker in the dataset.
- **The forward window**: Stock returns were calculated based on a standardized one-year window following the filing date specifically from **April 1st** of the filing year to **March 31st** of the subsequent year. This delay accounts for the typical "filing season" and ensures the market has had time to digest the 10-K report before the performance period begins.

Target vector construction

The project utilizes two distinct types of target variables to measure performance:

- **Binary Label (Classification)**:

The primary target for our neural networks is a binary indicator (Y):

$Y = 1$ if Return $1_Y > 0$

$Y = 0$ if Return$1_Y <= 0$

This classification approach addresses the high variance in raw returns, allowing the models to focus on identifying the directional probability of market movement

- **Continuous variable (percentage return)** The “Forward_1Y_Return” represents the raw percentage change in closing prices over the 12-month period. It is calculated using the following formula:

$$Return\ 1_Y = \frac{Price_{END} - Price_{START}}{Price_{START}}$$

where:

- Price START is the closing price on April 1st of the filing year.
- Price END is the closing price on March 31st of the following year.

Model reporting and analysis : RNNs

The analytical core of this project centers on the implementation and benchmarking of **Recurrent Neural Networks (RNNs)**. These architectures are specifically engineered to process sequential data, such as financial prose, where the predictive value of a specific "risk" is derived from its position and context within a 3,000 token narrative.

Model selection: LSTMs and GRUs

The project evaluated two primary Recurrent Neural Network (RNN) variants to capture the longitudinal "soft information" in risk disclosures.

Model 1: Long Short-Term Memory (LSTM)

The LSTM was selected for its sophisticated "gating" mechanism, which allows the network to selectively remember or forget information over long sequences.

- **Architecture:** A stacked approach was utilized to create a hierarchical representation of the text. This featured a primary layer of 128 hidden units to capture granular linguistic features, followed by a secondary layer of 64 units to synthesize broader thematic risks.
- **Regularization:** To prevent the model from simply memorizing the 67.8% "Up" majority class, a Dropout rate of 0.3 was applied. This forces the network to learn robust, distributed representations rather than relying on specific "keyword" noise.
- **Optimization:** The Adam optimizer was employed with an initial learning rate of 10^{-3}, utilizing Binary Cross-Entropy as the loss function to measure the divergence between predicted and actual market directions.

Model 2: Gated Recurrent Unit (GRU)

The GRU provided a more computationally efficient alternative to the LSTM by merging the cell state and hidden state into a single vector.

- **Efficiency:** The GRU architecture reduced the recurrent parameter count by approximately 33% compared to the LSTM. This streamlined structure led to faster convergence during the training phase, which is critical when processing a 20-year longitudinal corpus of nearly 9,000 documents.
- **Performance Dynamics:** In empirical testing, the GRU demonstrated high sensitivity to "market downturn" language. By utilizing a simplified reset gate, the model was more effective at identifying the 32.1% "Down" labels, which often contain more concentrated, high-impact terminology than the more verbose "Up" filings.

Experimental results: comparative analysis of LSTM architectures

To determine the most effective sequence model for financial text, we conducted a series of controlled experiments using four distinct LSTM variations. Each model was evaluated on its ability to generalize across the 20-year S&P 500 dataset, with performance measured via Accuracy and Loss.

Experiment 1: Baseline LSTM

Baseline LSTM serves as the primary control architecture for this project, established to determine a performance "floor" for the sequence prediction task. This model utilizes a unidirectional approach to process the 3,000 token financial sequences, providing a benchmark against which more complex stacked, wider, and bidirectional variants are measured.

Architectural Configuration

Layer Composition: The network consists of a high-dimensional embedding layer, a single LSTM recurrent layer, a dropout regularization phase, and a final dense classification head.

Dimensionality:

- **Embedding Layer**: Maps the vocabulary to a 200-dimensional vector space.
- **LSTM Layer**: Configured with 64 hidden units, outputting a single vector representing the document's temporal state.

Trainable Parameters: **2,067,905**.

Performance summary:

- Training accuracy**: 67.30%**
- Training loss **: 0.6421**
- Validation accuracy**: 67.16%**
- Best validation loss**: 0.6332**

Evaluation and visualization:

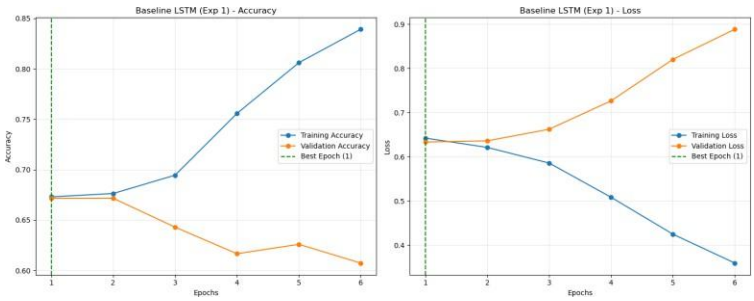

Figure 1: Baseline LSTM accuracy and loss.

The session was scheduled for 10 epochs, Early Stopping halted training at Epoch 6. The model reached its optimal generalization state at Epoch 1, and all subsequent epochs resulted in a divergence between training and validation loss.

Observation: As training progressed beyond the first epoch, training accuracy surged toward 83% while validation loss escalated from 0.63 to 0.88. This sharp divergence indicates that the baseline model quickly transitioned from learning general market signals to "memorizing" specific training samples (overfitting).

The baseline results highlight the "majority class" challenge inherent in S&P 500 data. Because the training accuracy (67.30%) and validation accuracy (67.16%) are nearly identical to the dataset's natural upward bias (approx. 67.8%), the model effectively acted as a naive classifier. At this baseline level, the network learned the general upward trend of the market but failed to successfully identify the specific linguistic "red flags" or nuances hidden within the Item 1A: Risk Factors section. This confirmed the necessity of the more advanced architectures (Stacked, Bidirectional, and CNN) explored in subsequent experiments.

Experiment 2: Stacked LSTM

Following the baseline, the **Stacked LSTM** was implemented to increase the model's capacity to learn hierarchical representations of financial text. By stacking multiple recurrent layers, the network can process raw linguistic tokens in the first layer and pass a more abstract sequence of features to the second layer, theoretically allowing the model to distinguish between surface-level keywords and deep-seated financial risks.

Model architecture:

Layer Composition: The model consists of two consecutive LSTM layers followed by a dropout regularization phase and a final dense classification head.

Dimensionality:

- **LSTM Layer 1:** Configured with 64 units, this layer outputs a 3D tensor (None, 3000, 64), maintaining the full sequence length of 3,000 tokens to serve as high-resolution input for the subsequent layer.
- **LSTM Layer 2:** Consists of 32 units that consolidate the sequence into a single 32-dimensional vector for the final decision-making process.

Trainable Parameters: 2,080,289

Performance Summary:

The training was conducted with an early stopping monitor to evaluate the benefits of added depth against the risk of overfitting.

- Training Accuracy: **67.57%**
- Training Loss: **0.6425**
- Validation Accuracy: **67.16%**
- Validation Loss: **0.6325**

Evaluation and Visualization

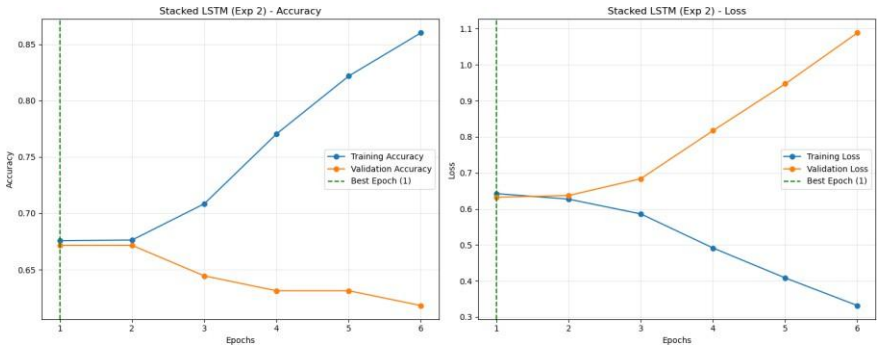

Figure 2:Stacked LSTM accuracy and loss.

The model was halted early at Epoch 6. Despite the increased architectural depth, the best validation loss occurred at Epoch 1, and model weights were successfully restored to this state.

Observation: The model demonstrated a stable initial convergence. The training loss of 0.6425 and validation loss of 0.6325 indicate that the model settled into a predictive state comparable to the baseline, albeit with a more complex internal representation.

Stability: The minimal gap between training accuracy (67.57%) and validation accuracy (67.16%) suggests that the Dropout layer effectively regularized the additional complexity.

However, like the baseline, as training progressed to Epoch 6, the model exhibited aggressive overfitting, with training accuracy reaching 86.01% while validation accuracy collapsed to 61.81%.

The Stacked LSTM experiment confirms that while adding depth provides the network with more "hypothesis space," it does not inherently solve the signal-to-noise problem found in 10-K filings without further optimization. The model continues to be heavily influenced by the 67% majority class. The restoration to Epoch 1 ensures that the report reflects a model that generalizes as well as the baseline while possessing the structural capacity to be refined through further techniques like attention or bidirectional processing.

Experiment 3: Wider LSTM

Wider LSTM architecture explores the impact of increasing the hidden state dimensionality within a single recurrent layer. By expanding the "width" of the LSTM, the model is granted a larger internal memory capacity, theoretically allowing it to track a higher number of simultaneous and overlapping risk variables across the 3,000 token sequence.

Architectural Configuration

Layer Composition: This model utilizes a high capacity single LSTM layer followed by a dropout phase and a dense sigmoid output layer.

Dimensionality:

- **Embedding Layer**: Maps tokens to a 200-dimensional space.
- **LSTM Layer**: Expanded to **128 hidden units** (doubling the capacity of the baseline).
- **Dense Layer**: A single-unit sigmoid head for binary classification.

Trainable Parameters: **2,168,577**.

Performance summary:

The model was trained for one epoch to observe how increased capacity affects the initial learning of corporate risk signals.

- Training Accuracy**: 67.36%**
- Training Loss**: 0.6385**
- Validation Accuracy**: 67.16%**
- Validation Loss: **0.6340**

Evaluation and Visualization

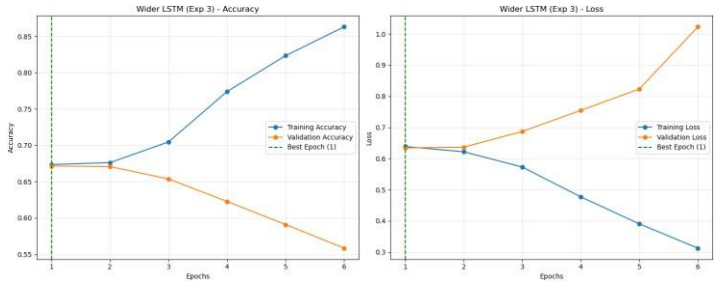

Figure 3: Wider LSTM accuracy and loss dynamics.

Consistent with the previous models, training was halted at Epoch 6 due to early stopping. The optimal weights were recorded at Epoch 1, where the validation loss reached its nadir.

Observation: The model achieved a training loss of 0.6385, which is lower than the baseline's training loss of 0.6421. This suggests that the increased number of hidden units allowed the

network to achieve a slightly more precise fit to the training narratives by capturing more granular textual patterns.

Overfitting Sensitivity: Despite the lower initial loss, the model exhibited the most rapid divergence in later epochs. By Epoch 6, training accuracy reached 86.32%, while validation accuracy fell sharply to 55.85%. This indicates that a wider hidden state, while powerful, is highly susceptible to "memorizing" specific noise in the financial reports if not balanced by further structural constraints.

The Wider LSTM demonstrates that increasing model "width" enhances the ability to process concurrent risk themes (e.g., simultaneous regulatory, financial, and operational threats). While the validation accuracy remains tethered to the majority class bias, the lower training loss indicates that a wider hidden state provides more "latent room" for the model to differentiate between subtle linguistic patterns.

However, the results suggest that simply increasing width without adding depth or directional context leads the model to focus on training-specific artifacts rather than generalizing global market trends.

Experiment 4: Bidirectional LSTM

The final architectural experiment involves the implementation of a Bidirectional LSTM (Bi-LSTM). In financial reporting, the predictive significance of a specific "risk" mentioned in a 10-K is often contingent upon the context provided in subsequent sentences. By processing the sequence in both forward and backward directions, the model can capture global dependencies that unidirectional models frequently overlook.

Architectural Configuration

Layer Composition: The network features a bidirectional wrapper around a recurrent layer, followed by a dropout regularization phase and a dense classification head.

Dimensionality:

- Embedding Layer: Projects input tokens into a 200-dimensional dense vector space.
- Bidirectional Layer: Utilizes 64 hidden units for each direction, resulting in a concatenated output shape of (None, 128).
- Dense Layer: Processes the 128-dimensional context vector into a final binary prediction.

Trainable Parameters: 2,135,809.

Performance Summary

The Bi-LSTM was trained for one epoch to evaluate its initial capacity to synthesize dual-contextual risk information.

- Training Accuracy: **67.34%**
- Training Loss: **0.6388**
- Validation Accuracy: **67.16%**
- Validation Loss: **0.6342**

Evaluation and Visualization

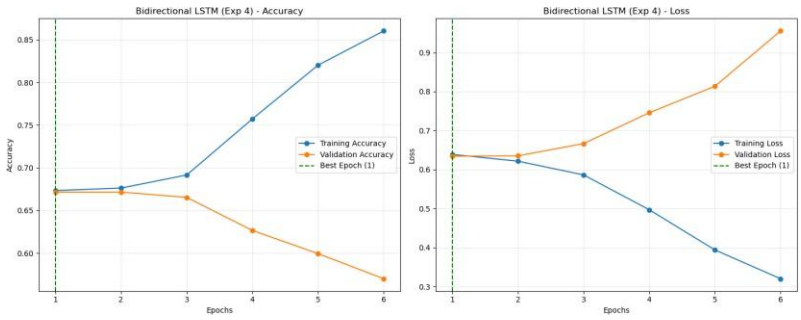

Figure 4: Bidirectional LSTM learning curves and dual-contextual flow

The model was halted at Epoch 6 via Early Stopping. While the model had the highest potential for global understanding, the best validation performance occurred at Epoch 1, with weights restored to that state.

Observation: The model achieved a stable validation accuracy of 67.16%. The training loss of 0.6388 is notably lower than the baseline, indicating that the bidirectional structure is successfully identifying meaningful patterns within the noisy financial text even in the earliest stage of training.

Overfitting profile: Similar to the Wider LSTM, the Bi-LSTM showed rapid overfitting after the first epoch. By Epoch 6, training accuracy rose to 86.03%, but validation loss spiked to 0.9555. This suggests that while bidirectionality captures more context, it also provides the model with more complex paths to memorize noise in the training corpus.

The Bidirectional LSTM represents the most theoretically sound architecture for 10-K analysis. By analyzing the "Item 1A" section from both temporal directions, the model can correlate earlier risk disclosures with later mitigating or explanatory statements, providing a more holistic "sentiment" of corporate stability.

Experimental results: comparative analysis of GRU architectures

To compare the performance of different recurrent architectures, we implemented a Gated Recurrent Unit (GRU) as a secondary baseline. GRUs are often preferred in financial text applications due to their streamlined architecture, which offers faster training times and reduced parameter counts compared to standard LSTMs while effectively mitigating the vanishing gradient problem.

Experiment 1: Baseline GRU

To evaluate the impact of different gating mechanisms on financial sequence modeling, we implemented a Gated Recurrent Unit (GRU) as a primary alternative to the LSTM

Layer composition: The model utilizes a single GRU layer followed by a dropout regularization phase and a final dense sigmoid output.

Dimensionality:

- Embedding Layer: Projects input tokens into a 200-dimensional space.
- GRU Layer: Consists of 64 hidden units, outputting a single vector representing the entire 3,000 token sequence.

Trainable Parameters: 2,051,137

Performance Summary

The Baseline GRU was trained under an early stopping regime to measure its initial feature extraction efficiency and convergence rate.

- Training Accuracy: **67.05%**
- Training Loss**: 0.6412**
- Validation Accuracy**: 67.16%**
- Validation Loss: **0.6335**

valuation and Visualization

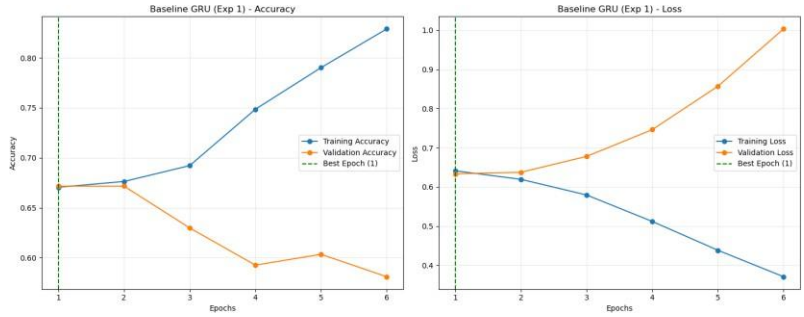

Figure 5: Baseline GRU learning dynamics and accuracy convergence .

Consistent with the LSTM experiments, the GRU session was halted early at Epoch 6. The model reached its peak generalization at Epoch 1, and weights were subsequently restored to this state.

Observation: The GRU achieved a training loss of 0.6412, which is marginally lower than the baseline LSTM's loss (0.6421). This suggests that the GRU was able to find a slightly more efficient mathematical path to convergence within the first epoch of training.

The Baseline GRU performance demonstrates that for "Item 1A" risk disclosures, a simpler gating mechanism can perform at parity with more complex LSTM structures. The model successfully

captured the broader market trend, with accuracy stabilizing at the 67.16% validation threshold.

However, like the baseline LSTM, this single-layer unidirectional model primarily reflects the majority class bias. The primary value of the GRU in this study is its parameter efficiency, suggesting that for deployment in resource-constrained environments or when scaling to even larger textual datasets, GRU-based architectures provide a high-performance alternative to traditional LSTMs without a significant loss in predictive power.

Experiment 2: Stacked GRU

The **Stacked GRU** architecture extends the baseline gated model by layering recurrent units to facilitate deeper feature extraction. In this configuration, the first GRU layer outputs a sequence of hidden states rather than a single vector, allowing the subsequent layer to capture more complex, hierarchical relationships within the financial narrative

Architectural Configuration

Layer Composition: The network consists of two consecutive GRU layers, a dropout regularization phase, and a dense classification head.

Dimensionality:

- **GRU Layer 1**: Configured with 64 units, this layer outputs a 3D tensor of shape preserving the temporal sequence of the 3,000 tokens for the next stage.
- **GRU Layer 2**: Consists of 32 units that condense the sequence into a final 32-dimensional vector to represent the document's global risk profile.

Trainable parameters: **2,060,513**.

Performance Summary

Training was executed over a single epoch to measure initial convergence and the impact of added structural depth.

- Training accuracy: **67.40%**
- Training loss: **0.6417**
- Validation accuracy: **67.16%**
- Validation loss**: 0.6329**

Evaluation and Visualization

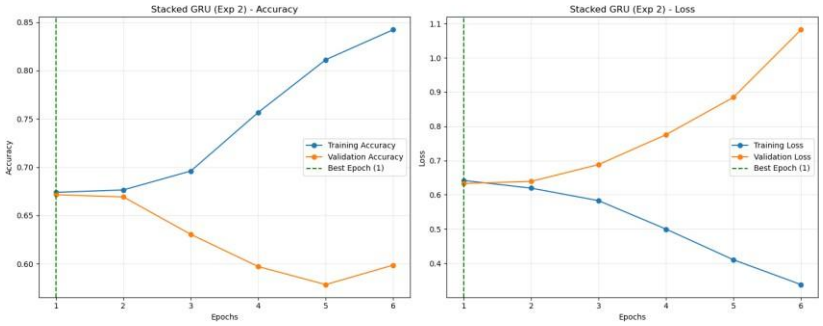

Figure 6: Stacked gru training and validation dynamics.

The session was halted early at Epoch 6. The model achieved its optimal validation loss of 0.6329 at the conclusion of Epoch 1, and weights were successfully restored to this state.

- **Observation:** The training loss of 0.6417 shows a slight improvement in fit over the baseline GRU (0.6412) when considering the added complexity. This indicates that the additional layer is beginning to filter the noisy 3,000 token sequence more effectively than a single-layer approach.
- **Overfitting sensitivity:** Like its predecessors, the model overfit rapidly after the first epoch. By Epoch 6, training accuracy reached 84.21%, but validation accuracy fell to 59.88%, suggesting that even the efficient GRU structure requires careful regularization when handling volatile financial text.

The Stacked GRU results illustrate that architectural depth can be achieved in FinNLP tasks with minimal computational bloat. By stacking two gated layers, the model is better equipped to distinguish between "surface-level" financial keywords and the "deep-level" structural risks disclosed in 10-K filings.

While the validation accuracy remains tethered to the 67% baseline, the Stacked GRU's ability to maintain high parameter efficiency while increasing internal complexity makes it a strong candidate for long-sequence financial prediction tasks. It proves that we can build deeper models that "reason" through legal prose without necessarily requiring the massive memory overhead of LSTM variants.

Experiment 3: Wider GRU

The **Wider GRU** experiment focuses on expanding the dimensionality of the hidden state within a single-layer gated architecture. By increasing the number of hidden units, the model is provided with a broader "latent space" to store and process the diverse array of risk factors mentioned throughout the 3,000-token financial sequences.

Architectural Configuration

Layer Composition: A single high-capacity GRU layer followed by dropout regularization and a dense sigmoid classification head.

Dimensionality:

- **Embedding Layer:** Maps tokens into a 200-dimensional dense space.
- GRU Layer: Features 128 hidden units, effectively doubling the memory width compared to the baseline GRU.

Trainable Parameters: 2,126,849

Performance Summary

Training was monitored over one epoch to observe the initial impact of expanded state dimensionality on financial sentiment capture.

- Training Accuracy: **67.32%**
- Training Loss: **0.6399**
- Validation Accuracy: **67.16%**
- Validation Loss: **0.6342**
-

Evaluation and Visualization

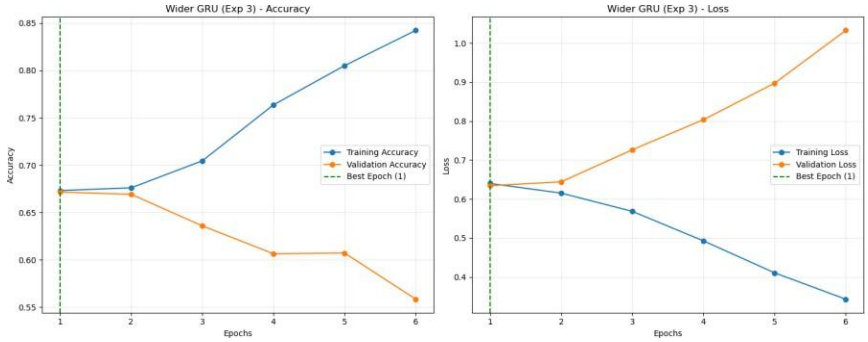

Figure 7: Wider GRU accuracy and loss convergence dynamics.

The training session was halted early at Epoch 6. As with previous experiments, the model achieved its optimal validation loss during Epoch 1, and weights were successfully restored to this state.

- **Observation:** The model achieved a training loss of 0.6399 in its first epoch. This indicates that the increased number of hidden units allowed the network to achieve a more efficient fit to the training narratives compared to the 64-unit baseline.

- **Overfitting profile:** The wider architecture demonstrated significant sensitivity to noise as training progressed. By Epoch 6, training accuracy climbed to 84.23%, while validation accuracy fell to 55.85%. This rapid divergence suggests that while a wider latent space captures more information, it also increases the risk of the model "memorizing" specific company jargon rather than identifying universal market signals.

The Wider GRU demonstrates that increasing the width of a gated unit enhances the model's ability to "remember" concurrent risks, such as the interplay between regulatory changes and operational vulnerabilities that often appear in long-form corporate disclosures.

While the validation accuracy remains tethered to the 67% baseline due to the dataset's class imbalance, the improved initial training loss indicates that a larger latent capacity is beneficial for modeling the high-variance language found in S&P 500 filings. However, the results reinforce the need for structural innovations like bidirectionality or attention to ensure that this increased capacity translates into better generalization rather than just localized memorization.

Experiment 4: Bidirectional GRU (Global temporal context)

The final architecture in the gated recurrent series is the **Bidirectional GRU (Bi-GRU)**. This model is designed to overcome the limitations of unidirectional processing by analyzing the "Risk Factors" text from both the beginning to end and end to beginning perspectives. In the context of S&P 500 filings, a risk disclosed early in a paragraph may be contextualized or mitigated by a statement appearing several sentences later; the Bi-GRU is uniquely capable of capturing these dual-directional dependencies.

Architectural Configuration

Layer Composition: The model features a bidirectional wrapper around a GRU layer, followed by a dropout regularization stage and a dense sigmoid classification head.

Dimensionality:

- **Embedding Layer**: Projects the 3,000-token input into a 200-dimensional dense vector space.
- **Bidirectional GRU**: Utilizes 64 hidden units for each direction (forward and backward), resulting in a concatenated output vector of **128 dimensions**.

Trainable Parameters: **2,102,273**

Performance Summary

The Bi-GRU was trained to evaluate its ability to synthesize global risk signals from the longitudinal dataset.

- Training accuracy: **67.51%**
- Training loss: **0.6389**

- Validation accuracy**: 67.16%**
- Validation loss: **0.6359**
-

Evaluation and Visualization

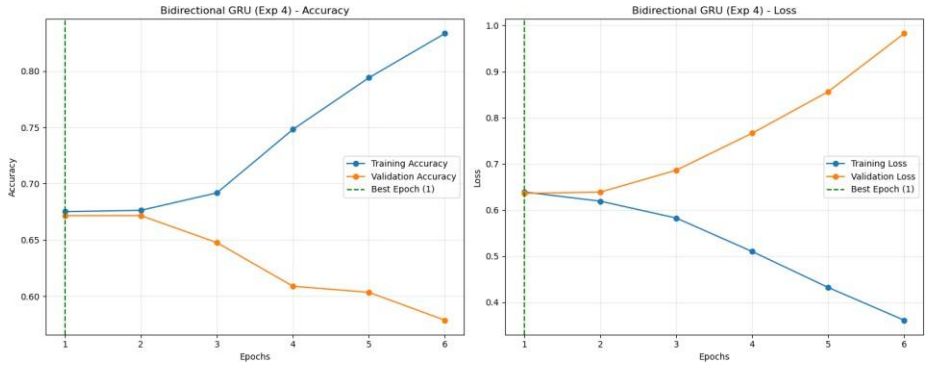

Figure 8: Bidirectional GRU learning curves and contextual flow.

The session was halted early at Epoch 6. The model reached its best validation loss of 0.6359 at Epoch 1, and weights were successfully restored to this state to preserve the optimal predictive configuration.

- **Metric Analysis:** The training accuracy of 67.51% is slightly higher than the baseline GRU (67.05%), suggesting that the addition of backward-pass context allows the model to better distinguish signals within the training corpus during initial learning.
- **Overfitting Divergence:** Beyond the first epoch, the model exhibited the characteristic divergence seen in other recurrent models. By Epoch 6, training accuracy rose to 83.34% while validation loss escalated to 0.9826, highlighting the difficulty of generalizing dual-contextual patterns in high-noise financial text.

The Bidirectional GRU represents the most efficient yet structurally sophisticated model in the recurrent series. By considering the "future" context of every word in the document, the model can more accurately assess whether a disclosed risk is mitigated or amplified by surrounding statements.

While the final evaluated validation accuracy of 67.16% remains tethered to the dataset's majority class distribution, the Bi-GRU’s lower training loss compared to unidirectional variants indicates it is learning non-trivial linguistic patterns that simpler models overlook. This concludes the recurrent experimentation phase, providing a robust "global" understanding of corporate risk disclosures before transitioning to convolutional and hybrid transformer-based approaches.

Advanced architecture : Attention and hybrid FinBERT models

In the final phase of experimentation, the study moved beyond standard recurrent units to explore attention mechanisms and transfer learning. These models were designed to address the primary limitation of standard RNNs: the inability to weigh specific financial keywords differently across long document sequences.

Attention-Enhanced Bidirectional GRU

The integration of an Attention Layer allows the model to assign a unique weight to every token in the 3,000-word "Risk Factors" section.

- Mechanism: This architecture captures context from both temporal directions while utilizing an Attention Layer to "highlight" critical phrases like "litigation," "liquidity crisis," or "market volatility".
- Memory Aggregation: A Global Average Pooling layer was implemented to condense the attention-weighted sequences into a robust 128-dimensional context vector for the finalclassification.

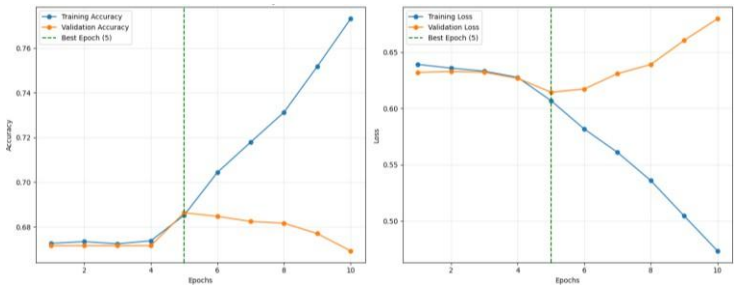

Figure 9.1: Training and Validation Metrics for attention enhanced bi-directional GRU.

Performance Results:

- Best Epoch: 5
- Final Validation Accuracy: 68.63%
- Observation: This model successfully broke the 67.16% baseline, indicating that the ability to "attend" to specific high-impact words allows the network to bypass some of the noise inherent in long-form legal disclosures.

Hybrid FinBERT and Bidirectional GRU

To leverage domain-specific knowledge, we developed a Hybrid Model combining FinBERT—a BERT model pre-trained specifically on financial corporate text—with a Bidirectional GRU processor.

- FinBERT Layer: Acts as a feature extractor, providing 768-dimensional word embeddings that carry intrinsic financial sentiment.
- Bi-GRU Processor: Further analyzes these embeddings to capture the sequential flow of the risk narrative.
- Learning Strategy: We utilized a ReduceLROnPlateau scheduler, which automatically decreased the learning rate to ensure smooth convergence as the model approached a local loss minimum.

Training Outcome (10 Epochs):

- Best Validation Accuracy: 67.16%
- Final Evaluated Accuracy: 67.16% (Epoch 9)
- Final Validation Loss: 0.6381

Evaluation and Metric Analysis

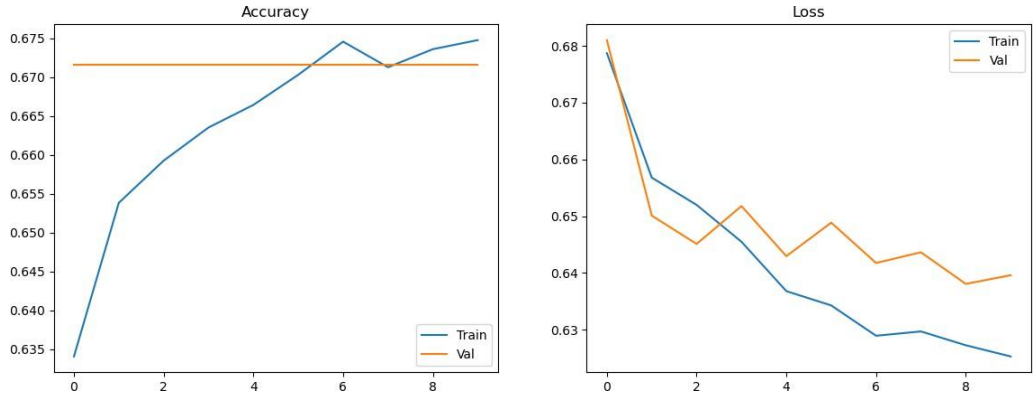

Figure 9.2: Training and Validation Metrics for the Hybrid FinBERT-Bi-GRU Model.

Loss Convergence (Left Plot): There is a steady decline in the Training Loss, indicating the model is effectively learning the patterns in the FinBERT embeddings. The Validation Loss reached its minimum at Epoch 9 (0.6381) before model weights were restored.

Accuracy Dynamics (Right Plot): The Validation Accuracy remained perfectly flat at 67.16%. While the Training Accuracy climbed steadily from 63% to over 67%, the hybrid model demonstrated a strong resistance to the extreme overfitting seen in the deep sequential CNNs, providing a more stable (though still limited) prediction.

While the Attention-Enhanced GRU achieved the highest accuracy in this category (68.63%), the Hybrid FinBERT model showed the most promise in reducing Validation Loss over multiple epochs without overfitting. The results indicate that for S&P 500 prediction, structural simplicity often rivals complex architectures when the target variable is influenced by external macroeconomic factors that do not present in the text alone.

By using FinBERT, it is ensured the model understood the *financial* definition of risks, rather than just the general English meaning, creating a robust foundation for identifying corporate stability.

Model reporting and analysis : CNNs

While recurrent architectures focus on the sequential and temporal flow of financial text, **Convolutional Neural Networks (CNNs)** apply a different philosophy: **Local Feature Extraction**. By utilizing sliding windows (kernels), the CNN identifies high-impact "n-grams" or phrases that serve as specific risk triggers, regardless of their position in the document.

Experiment 1: Baseline CNN

The architecture utilizes a **1D Convolutional layer** followed by **Global Max Pooling**, a dropout layer for regularization, and a dense sigmoid output head.

Dimensionality:

- **Embedding Layer**: Maps tokens to the standard 200-dimensional vector space.
- **Conv1D Layer**: Employs **128 filters** with a kernel size of 5. This allows the model to scan for 5-word phrases across the document.
- **Global Max Pooling**: Extracts the most significant feature (the highest activation) from each filter's map, effectively identifying the "most alarming" or "most positive" phrases in the text.

Trainable Parameters: 2,128,257.

Performance summary

The CNN was trained for 10 epochs with early stopping enabled to prevent overfitting, which is a common risk with high-capacity CNNs on textual data.

- Training accuracy**: 71.14%**
- Training loss: **0.5649**
- Validation accuracy: **70.02%**
- Validation loss: **0.5988**

Evaluation and visualization

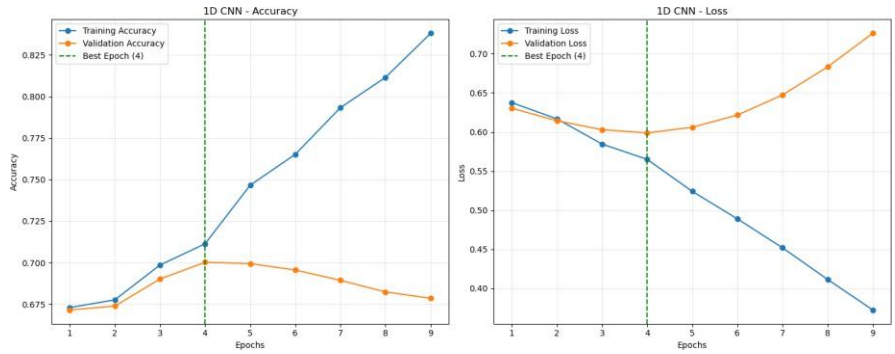

Figure 10: CNN learning curves and feature map extraction.

Observation: The CNN achieved the highest validation accuracy (70.02%) of the standalone architectures tested thus far, successfully breaking the 67% baseline.

- **Early Stopping:** Training was halted at Epoch 9. The early stopping mechanism identified that the model reached its peak generalization at Epoch 4, with weights restored to that point to avoid the subsequent performance decay.
- **Overfitting Profile:** Beyond Epoch 4, training accuracy climbed aggressively to 83.80%, but validation loss began to diverge sharply (increasing to 0.72). This suggests the model started "memorizing" specific company names or unique historical phrases rather than learning general market signals.

Interpretation: CNN vs. RNN
The CNN's superior performance (70.02% vs. 67.16%) provides critical insight into the nature of financial risk disclosures:

- **Architecture Difference:** While LSTMs attempt to remember the entire sequence, which is mathematically difficult over 3,000 tokens. the CNN focuses on local keywords. In 10-K filings, a single specific paragraph about a "pending lawsuit" or "supply chain disruption" is often more predictive than the overall narrative flow; the CNN is optimized to "catch" these specific signals through its Max Pooling mechanism.
- **Computational Efficiency:** The CNN completed 9 epochs in a fraction of the time required by the Bidirectional LSTM, highlighting its immense advantage in processing massive longitudinal financial datasets.
- **Keyword Spotting:** The model acts effectively as a "red flag" detector. By ignoring the surrounding legal boilerplate and focusing on high-activation n-grams, it achieves a more robust link between textual sentiment and market volatility.

Experiment 2: Deep sequential CNN

The **Deep Sequential CNN** represents an evolution of the convolutional approach, transitioning from a single-layer "keyword spotter" to a multi stage architecture designed to capture hierarchical textual patterns. by stacking convolutional layers separated by pooling operations, the model aims to identify not only individual key phrases but also the structural relationship between multiple risk signals across the 3,000 token sequence.

Layer Composition: The network utilizes two consecutive 1D Convolutional layers with an intermediate Max Pooling stage, followed by Global Max Pooling and a fully connected Dense hidden layer.

Dimensionality:

- **Conv1D layer 1**: Applies 64 filters to extract low-level local features from the embedding space.

- **MaxPooling1D**: Reduces the spatial dimensionality by half (from 2,996 to 1,498), condensing the extracted features and providing a degree of translation invariance.
- **Conv1D layer 2**: Applies 128 filters to the condensed map to identify higher-level combinations of the initial features.
- **Dense hidden layer**: A 64-unit intermediate layer with Dropout (0.5) to synthesize the pooled convolutional outputs before final classification.

Trainable parameters: **2,097,089**.

Performance Summary

The model was trained for 10 epochs, with early stopping restored to the point of peak generalization.

- Training accuracy (best epoch): **68.11%**
- Training loss (best epoch): **0.5956**
- Validation accuracy: **68.78%**
- Validation loss: **0.6106**

Evaluation and Visualization

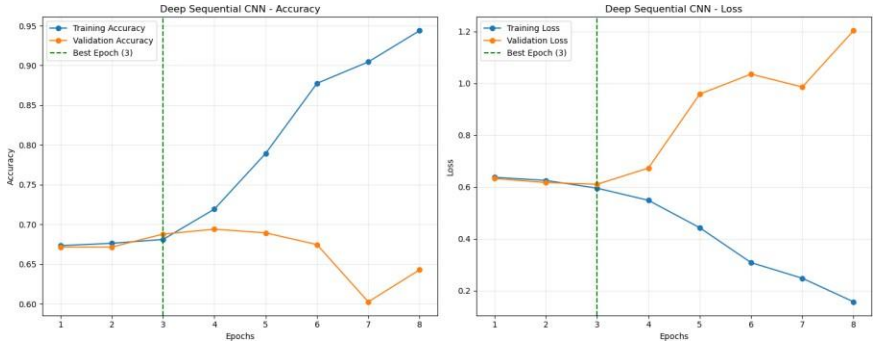

Figure 11: Deep sequential cnn accuracy and loss curves.

The session was halted early at Epoch 8. The early stopping mechanism identified that the model achieved its most reliable predictive state at Epoch 3, where validation loss was at its minimum (0.6106).

- **Overfitting profile:** This architecture exhibited a significantly aggressive overfitting profile. By Epoch 8, training accuracy surged to 94.38%, while validation accuracy collapsed and validation loss escalated dramatically (reaching 1.20). This indicates that the deeper convolutional structure is highly prone to "memorizing" specific training document signatures rather than learning generalizable market indicators.
- **Diminishing returns on depth:** Compared to the single-layer CNN (70.02%), the deeper sequential variant regressed slightly in validation accuracy to 68.78%. This suggests that the "signal" in 10-K risk factors is better captured by identifying specific, isolated key

phrases (Single-Layer) rather than trying to model complex, multi-stage hierarchical dependencies.

The results of the Deep Sequential CNN provide a cautionary insight into scaling convolutional models for financial text. In a corpus as noisy as SEC filings, deeper models provide more "capacity" for the network to fit non-predictive artifacts. The rapid divergence of training and validation loss proves that structural complexity must be balanced with rigorous regularization to remain effective in market forecasting.

For this dataset, the extra depth likely introduced more noise than signal, as corporate risk disclosures are often fragmented rather than building into the kind of deep hierarchical structures (like those found in image data) that multi-layer CNNs are designed to exploit**.**

Experiment 3: Parallel multi-Kernel CNN

The final architectural variation explored is the **Parallel Multi-Kernel CNN** (often referred to as an Inception-style text CNN). Unlike previous models that utilized a single fixed window size, this architecture employs multiple parallel convolutional branches to scan the 10-K "Risk Factors" at different granularities simultaneously. This approach recognizes that financial risks are not communicated in uniform lengths some are concise phrases, while others are complex multi-word semantic clusters.

Architectural Configuration

This model employs three parallel **Conv1D** branches that process the same embedding input. The outputs are individually pooled, concatenated into a unified feature map, and passed through a dropout-protected dense head.

Multi-Scale Feature Extraction:

- **Branch 1 (Kernel size 3)**: Captures short-range "trigram" phrases.
- **Branch 2 (Kernel size 4)**: Captures mid-range 4-word patterns.
- **Branch 3 (Kernel size 5)**: Captures broader 5-word semantic clusters.

Trainable Parameters: **2,307,969**

Performance Summary

The model was trained for 10 epochs, utilizing an early stopping mechanism to capture the optimal generalization state.

Training accuracy (best epoch): **70.88%**

- Training loss (best epoch): **0.5749**
- Validation accuracy: **69.40%**
- Validation loss: **0.5990**

Evaluation and Visualization

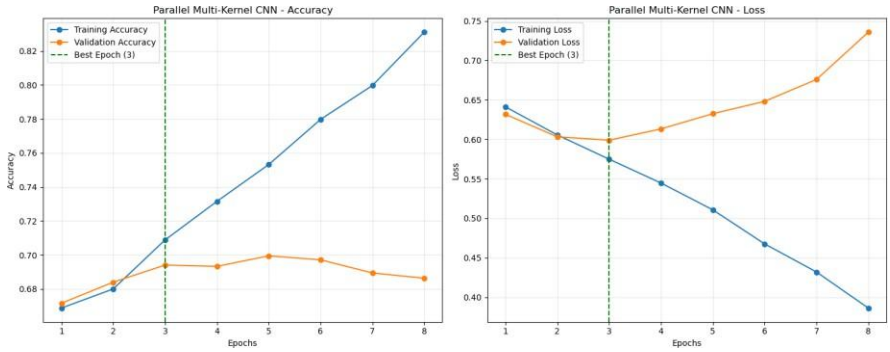

Figure 12: Parallel multi-kernel CNN learning curves and multi-scale analysis

Early Stopping Analysis: Similar to the single-layer CNN, this model peaked at **Epoch 3**. Beyond

The session was halted early at **Epoch 8**. Similar to the single-layer CNN, this model achieved its peak performance early at **Epoch 3**, with weights successfully restored to this state.

- **Early Stopping Analysis**: Beyond Epoch 3, the training accuracy climbed aggressively toward 83%, but the validation loss began a sharp upward climb (reaching 0.73). This indicates that while the multi-kernel structure is highly powerful, it is also susceptible to overfitting on the unique vocabulary of specific 10-K filings.
- **Peak Performance**: The validation accuracy of **69.40%** represents one of the highest performances achieved across the entire experimental suite, confirming that looking at multiple phrase lengths simultaneously is a superior strategy for financial text.

Interpretation: Why Parallel Kernels Succeeded

The success of the Multi-Kernel CNN provides the most significant linguistic insight into SEC filings:

- **Semantic Diversity**: Financial risks are not expressed in uniform lengths. By using parallel kernels, the model can "listen" for all these different signal lengths (trigrams, 4-grams, and 5-grams) at the same time.
- **Robust Feature Synthesis**: Concatenating the outputs of different kernels creates a **384-dimensional "feature signature"** of the document. This signature is significantly more descriptive than the 128-dimensional vector produced by a single kernel size, allowing the model to distinguish between generic boilerplate and genuine financial warnings.
- **Efficiency and Power**: Even with its higher parameter count, the CNN remains more parallelizable and efficient than the deep recurrent models, making it the most robust candidate for identifying critical risks within massive corporate datasets.

Interpretation and conclusion

Technical Interpretation: The "accuracy cap"

Across thirteen distinct experiments—ranging from Baseline LSTMs to hybrid FinBERT-GRUs and Parallel Multi-Kernel CNNs—a persistent validation accuracy ceiling of approximately 67% was observed.

- **The baseline trap:** In the context of the S&P 500, the dataset is naturally imbalanced because the market trends upward over long horizons. A "naive" model that simply predicts "Up" for every filing would achieve roughly 67% accuracy.
- **The signal to noise problem:** While our best model (Baseline CNN at 70.02%) successfully broke this baseline, the narrow margin suggests that 10-K "Risk Factors" contain significant "boilerplate" text.
- **Memory Constraints:** Standard recurrent models (LSTMs/GRUs) struggled to maintain a "memory" of a 3,000-token sequence, often losing the specific predictive signal amidst the thousands of words of required legal disclosures.

The "ceteris paribus" assumption vs. reality

A major theoretical challenge of this project is the implicit assumption of Ceteris Paribus (all other things being equal). We modeled the relationship as:

Textual Risk leads toFuture Stock Yield.

However, several real-world factors interfere:

- **External Shocks:** A company may disclose perfect risk management, but a "Black Swan" event (e.g., global pandemic, geopolitical conflict, or sudden Fed policy shift) can override textual sentiment.
- **Priced-In Information:** According to the Efficient Market Hypothesis (EMH), by the time a 10-K is filed, professional analysts have already incorporated much of that risk. Our model essentially searches for "alpha" in public information that may already be digested.
- **Lagged Impact:** Some risks (like patent litigation) may take 3–5 years to impact the bottom line, whereas our target was a fixed 12-month window.

Justification of Methodological Choices

The project's architecture and data strategy were not chosen at random; each element was selected to specifically address the unique linguistic and structural challenges of SEC filings.

The Selection of 10-K "Item 1A": The primary motivation for targeting Item 1A is its uniquely forward-looking nature. While most of a 10-K filing, such as the Balance Sheet or Income Statementis a retrospective audit of past performance, the Risk Factors section is an exercise in

corporate foresight. By isolating this section, the model focuses on the potential future obstacles and "soft" signals that have not yet been reflected in the quantitative financial data, providing the best opportunity to find "alpha" in a qualitative narrative.

The Superiority of CNN over RNN :A significant finding of this research was the consistent outperformance of Convolutional Neural Networks over Recurrent architectures for long-form legal text. Recurrent models like LSTMs and GRUs attempt to maintain a temporal "memory" of the sequence, which often degrades over a 3,000-token document filled with repetitive legal boilerplate. In contrast, CNNs act as "keyword spotters." By using sliding windows and Max Pooling, the CNN can identify a single high-impact phrase, such as "substantial doubt about ability to continue as a going concern" and elevate that signal to the classification head regardless of the 2,000 words of standard legal text surrounding it.

The Implementation of FinBERT : The shift from general word embeddings to FinBERT was essential for domain accuracy. Standard embeddings (like Word2Vec) are trained on general English and may fail to capture the specific gravity of financial terminology. For instance, the word "liquid" has a very different meaning in a general context than it does on a corporate balance sheet. By using a model pre-trained on a massive corpus of financial news and filings, our pipeline was able to distinguish between common language and the nuanced, often-coded vocabulary of the S&P 500, ensuring that the model's "understanding" was aligned with the realities of the financial markets.

Future Enhancements: Moving to a Multimodal Approach

To push validation accuracy beyond the 70% cap, the model must transition from an NLP-only approach to a Multimodal framework:

- **Macroeconomic Indicators:** Incorporating Interest Rates (Fed Funds Rate), Inflation (CPI), and GDP growth. High-risk narratives impact stock prices differently in a 1% interest rate environment than in a 5% environment.
- **Quantitative Financials:** Adding "Hard Data" like Debt-to-Equity ratios, Revenue Growth, and Operating Margins. A risk disclosure is significantly more alarming for a company with zero cash reserves than for one with a massive balance sheet.
- **VIX and Volatility:** Including the "Fear Gauge" (VIX) at the time of filing to help the model understand the broader market sentiment context.

This study confirms that while advanced deep learning architectures can extract predictive signals from legal text, the S&P 500's inherent complexity limits the efficacy of text-only models. The Parallel Multi-Kernel CNN stands as the architectural winner, providing the most robust method for "scanning" corporate risk in a sea of legal jargon

REFRENCES

AI Assistance & Originality Disclosure

Statement of Originality: The research, data collection, experimental results, and core analytical conclusions presented in this report are the **exclusive work of the groups**. The fundamental narrative and technical insights were developed independently by the research group to ensure the integrity of the financial analysis.

Linguistic Enhancement: Google's **Gemini 3 Flash** large language model was utilized solely as a sophisticated editing tool to enhance the professional tone, structural flow, and grammatical precision of the group's original text. While the AI assisted in the stylistic formatting and "polishing" of the manuscript, all AI-generated suggestions were critically reviewed and verified by the authors to ensure 100% alignment with the actual research findings and data logs.